



Research paper

Space-for-Time Substitution to study changes in flowering phenology in herbaceous species is most reliable for early-flowering species

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ABSTRACT

Phenological events in plants are shifting worldwide in response to changes in climate. Studying long-term shifts of phenological events in plants using historical data can lead to a better understanding of processes driving phenological changes and a basis for predicting potential changes in the future. However, conducting such Time-for-Time Substitutions (TFTS), i.e., learning from the past to predict the future, remains rare due to limited availability of long-term data. An alternative method for understanding phenological change is the Space-for-Time Substitution (SFTS), in which the temporal gradient is reflected by a spatial gradient.

We used historical phenological records of 17 herbaceous species from botanical gardens in Vienna covering a period of 150 years and data from 11 botanical gardens covering a spatial gradient to test the space-time equivalence. We fitted a linear-mixed effect model to test for differences between the two approaches (i.e., changes in time vs. changes in space) for analysing the associations between annual flowering onset dates and temperature. In addition, we examined the species-specific suitability of SFTS in regard to species' temporal niches.

We showed that flowering advanced on average by seven days for a temperature increase of 1 °C, irrespective of the gradient considered. At the species level, however, we found that the known pattern that late-flowering species are less temperature-sensitive was not found with the SFTS dataset. The space-time equivalence was better fulfilled for early-flowering species than for late-flowering ones.

This study contributes significantly to phenological research, as it demonstrated that the SFTS generally applied for herbaceous species. However, this method should always be used with caution for species-specific questions. In addition, it remains unclear to what extent the patterns found in botanical gardens are also found in less-controlled habitats.

Introduction

Since the 19th century, the Earth's climate has warmed significantly from relatively low temperatures during the Holocene (Marcott et al., 2013). Especially, during the past three decades, the average global temperature has risen considerably by 0.2 °C per decade (IPCC, 2014; Walsh et al., 2014). One of the most prominent effects of climate warming are shifts in the timing of phenological events in plants (Hassan et al., 2024; Ju et al., 2022; Richardson et al., 2013). Consequently,

phenology is an important indicator of climate change and an increasingly important research area (Parolini, 2022). Reports documenting phenological changes at global, regional, and local scales imply the holistic effects of rising temperatures (Berg et al., 2019; Piao et al., 2019; Song et al., 2021). Many studies have shown that plants have advanced spring phenology (Liu & Zhang, 2020; Menzel et al., 2006) and flowering onset (Menzel & Fabian, 1999; Miller-Rushing & Primack, 2008) while the onset of senescence is delayed (Huang et al., 2020).

The timing of phenological events such as onset of flowering results

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